

ID number:

Full Name:

Date:

Class 2 Quiz 2

Introduction to Object Oriented Programming – Fall 2025

Answer the questions in English, **closed-book, closed-notes, no electronics**. Time: 15 minutes

1. You can always successfully cast an instance of a superclass to a subclass. (True/False, 2 pts)

False

2. Sofa is a subclass of Furniture. How do you invoke an overridden toString() method in Furniture from within Sofa? (3 pts)

super.toString();

3. What is the role of this keyword in the code below? (3 pts)

```
public class Circle {  
    private double radius;  
    public Circle(double newRadius) {  
        radius = newRadius;  
    }  
    public Circle() {  
        this(1.0);  
    }  
}
```

Invoke another constructor in the same class

4. What is the output of the program below? Note: All classes are in the same package. (3 pts)

```
public class Test {  
    public static void main(String[] args) {  
        new Able();  
        new Baker();  
    }  
}  
class Able {  
    int i = 7;  
    public Able() {  
        setI(10);  
        System.out.println("i from Able is " + i);  
    }  
    public void setI(int i) {  
        this.i = i + 11;  
    }  
}  
class Baker extends Able {  
    public Baker() {  
        super();  
        System.out.println("i from Baker is " + i);  
    }  
}
```

Solution: i from Able is 21
i from Able is 21
i from Baker is 21